

IoT-Based Energy Monitoring System for Optimizing Power Consumption in University Facilities

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Abstract—This paper presents the development and implementation of an IoT-based Energy Monitoring System designed to optimize power consumption within university facilities. The system leverages real-time current and power readings, transmitted to a cloud-based Firebase database, to monitor energy usage across selected rooms at Caraga State University. Utilizing the ACS712 current sensor and ESP8266 Wi-Fi module, the system provides accurate and reliable data. Users can remotely control the power supply to various rooms via a web-based graphical user interface (GUI), enabling efficient energy management. The system's ability to switch electrical lines on or off remotely contributes to reducing unnecessary power consumption and lowering electricity costs. Performance tests demonstrate that the current sensor's accuracy is within $\pm 1.5\%$, confirming the system's reliability. While the current study excludes voltage sensors, it highlights the potential for future enhancements to further improve accuracy. This system is envisioned as a significant tool for promoting energy efficiency in educational institutions, with the potential for broader application in other sectors. The results indicate that the developed system can effectively monitor and control energy consumption, contributing to sustainable energy management practices.

Keywords—IoT-based Energy Management, ACS712 Current Sensor, ESP8266 Wi-Fi Module, Firebase Database, Real-Time Monitoring.

I. INTRODUCTION

A. Background

Electricity has become an indispensable component of modern society, shaping and driving the development of technology and infrastructure across the globe [1]-[6]. In the Philippines, as in many other countries, electricity plays a crucial role in daily life, powering everything from household appliances to industrial machinery. However, with the growing reliance on electrical energy comes the challenge of managing and optimizing its use, particularly in environments where large-scale consumption is inevitable, such as universities and other educational institutions [7]-[12].

The electricity sector is characterized by high costs and a growing demand that often outstrips supply, particularly during peak periods such as the summer months [13]-[16]. The Energy Regulatory Commission (ERC) oversees the regulation of power generation and distribution in the Philippines, but despite various initiatives aimed at reducing costs, electricity rates remain among the highest in Asia. For instance, the cost per kilowatt-hour (kWh) has seen a steady

increase over the past two decades, with prices climbing from \$0.11 USD in 2001 to \$0.33 USD in recent years. This trend has placed a significant financial burden on consumers, including educational institutions like Caraga State University (CSU), where electricity bills constitute a substantial portion of operational expenses.

The rising cost of electricity, coupled with the increasing demand for energy, underscores the importance of developing and implementing effective energy management strategies [17]-[19]. In response to this challenge, various technological solutions have been explored, including the use of smart grids, renewable energy sources, and energy-efficient appliances. Among these solutions, the Internet of Things (IoT) has emerged as a promising tool for enhancing energy management practices [20]-[25]. IoT technology enables real-time monitoring and control of electrical systems, allowing for more efficient use of energy and reducing wastage. By integrating IoT with energy management systems, institutions can gain better control over their energy consumption, ultimately leading to cost savings and more sustainable practices [26]-[31].

This paper focuses on the development and implementation of an IoT-based Energy Monitoring and Control System specifically designed for Caraga State University. The system aims to address the challenges of high energy costs and inefficient energy usage by providing a platform for real-time monitoring and control of electrical consumption in selected rooms and buildings on campus. Through the use of IoT technology, the system collects and analyzes data on energy usage, enabling university administrators to make informed decisions about energy management and to implement strategies that reduce overall consumption.

B. Problem Statement and Gaps

Despite the availability of various energy management technologies, many educational institutions in the Philippines, including Caraga State University, continue to struggle with high electricity costs and inefficient energy use. Existing systems often lack the ability to provide real-time data on energy consumption, making it difficult for administrators to identify and address areas of wastage. Moreover, traditional energy management systems are typically manual and reactive, meaning that energy-saving measures are often implemented only after high consumption has been recorded. This approach is not only inefficient but also costly, as it fails to prevent excessive energy use before it occurs.

The gaps in current energy management practices highlight the need for a more proactive and data-driven approach. An IoT-based system offers a solution by providing real-time data on energy consumption, allowing for immediate intervention when excessive usage is detected. Additionally, the ability to remotely control electrical systems means that energy-saving measures can be implemented quickly and efficiently, further reducing the potential for wastage. However, the implementation of such systems in educational institutions remains limited, particularly in the context of developing countries like the Philippines. This paper seeks to address this gap by presenting a case study of the development and deployment of an IoT-based energy management system at Caraga State University.

C. Goals and Objectives

The primary goal of this paper is to demonstrate the effectiveness of an IoT-based Energy Monitoring and Control System in optimizing energy consumption and reducing electricity costs at Caraga State University. To achieve this goal, the paper outlines several specific objectives:

1. Design and development of the system: The paper will detail the design and development process of the IoT-based energy management system, including the selection of hardware components, such as the ACS712 current sensor and the ESP8266 Wi-Fi module, and the integration of these components with a cloud-based database using Firebase.
2. Real-time monitoring and data collection: The system will be tested to ensure it can provide accurate, real-time data on energy consumption across selected rooms and buildings at Caraga State University. The paper will discuss the methods used to collect, store, and analyze this data, as well as the challenges encountered during the testing phase.

D. Significance

The significance of this paper lies in its contribution to the growing body of research on IoT-based energy management systems, particularly in the context of educational institutions in developing countries. By presenting a case study of the implementation of such a system at Caraga State University, the paper provides valuable insights into the challenges and benefits of adopting IoT technology for energy management. The findings of this study have the potential to inform the development of similar systems at other universities and institutions, both in the Philippines and beyond.

Moreover, the paper addresses a critical issue facing many educational institutions today: the rising cost of electricity. By demonstrating the effectiveness of an IoT-based system in reducing energy consumption and costs, the paper offers a practical solution that can help institutions manage their resources more efficiently. This is particularly important in the context of the Philippines, where high electricity costs can pose a significant financial burden on educational institutions.

In addition to its practical implications, this paper also contributes to the academic literature on IoT and energy management. While there has been considerable research on the use of IoT in various industries, there is still relatively little work on its application in the education sector, particularly in developing countries. This paper helps to fill that gap by providing a detailed analysis of the development and implementation of an IoT-based system in a university setting.

The insights gained from this study can be used to guide future research on IoT applications in energy management and other areas of university operations.

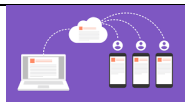
II. METHODOLOGY

The development and implementation of the IoT-based Energy Monitoring and Control System at Caraga State University followed a systematic methodology aimed at ensuring the system's reliability, accuracy, and efficiency. This section details the design process, selection of components, system architecture, development, testing, and deployment phases that were undertaken to create a functional and robust energy management solution.

A. System Design and Architecture

The system was designed to achieve real-time monitoring and remote control of energy consumption across selected rooms and buildings within the university. The core components of the system include the ACS712 current sensor, Arduino Uno microcontroller, ESP8266 Wi-Fi module, and Firebase cloud database (see Table 1).

TABLE I. CORE COMPONENTS OF THE DEVICE OR SYSTEM DEVELOPED.

Components	Reference images
ACS712 current sensor	
Arduino Uno microcontroller	
ESP8266 Wi-Fi module	
Firebase cloud database	

Each component was carefully selected to meet the specific requirements of the project, including accuracy, cost-effectiveness, and ease of integration.

The ACS712 current sensor was chosen for its ability to provide precise measurements of current flowing through the electrical circuits. This sensor is capable of detecting both AC and DC currents, with an accuracy of $\pm 1.5\%$, making it ideal for monitoring energy usage in real-time. The Arduino Uno microcontroller was selected due to its compatibility with the ACS712 sensor and the ESP8266 Wi-Fi module, as well as its ease of programming and widespread use in IoT projects. The microcontroller processes the data from the current sensor and sends it to the Wi-Fi module for transmission to the cloud database.

The ESP8266 Wi-Fi module was chosen for its reliability in establishing wireless connections and transmitting data to the cloud. This module connects to the university's wireless network and interfaces with the Firebase database, allowing real-time data collection and storage. Firebase, a cloud-based platform provided by Google, was selected as the database solution due to its real-time data synchronization capabilities, security features, and ease of integration with other

components of the system. The system architecture was designed to ensure seamless communication between the sensors, microcontroller, Wi-Fi module, and cloud database, enabling real-time monitoring and control of energy consumption.

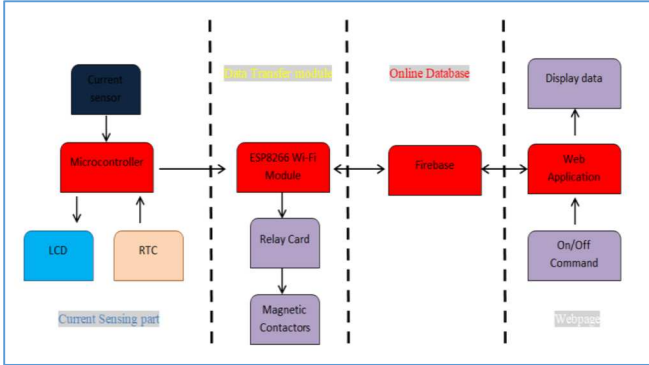


Fig. 1. System Architecture of the IoT-Based Energy Monitoring and Control System, illustrating the flow of data from the current sensing components to the web-based application through the microcontroller, Wi-Fi module, and Firebase online database.

The system architecture of the IoT-Based Energy Monitoring and Control System (see Fig. 1) is designed to facilitate real-time monitoring and remote control of energy consumption. The system begins with the current sensing components, which include the ACS712 current sensor connected to a microcontroller (Arduino Uno). The microcontroller processes the sensor data, converting it into a digital format. This data is then transmitted to a cloud-based Firebase database via the ESP8266 Wi-Fi module. The Firebase database stores the data, making it accessible to the web-based application, which displays the information in real-time. Users can interact with the web application to monitor energy usage and issue on/off commands for electrical systems, which are relayed back through the Wi-Fi module to the microcontroller, enabling precise control of the connected devices. This architecture ensures seamless data flow and control, optimizing energy management through a user-friendly interface.

B. System Development Process

The development process began with the design of the system's hardware and software components. The hardware design focused on integrating the ACS712 current sensor with the Arduino Uno microcontroller and ESP8266 Wi-Fi module. The sensor was connected to the microcontroller's analog input pins, allowing it to measure the current in the electrical circuits and convert it into a readable voltage signal. The microcontroller was programmed to process the sensor data, convert it into digital form, and transmit it to the Wi-Fi module.



Fig. 2. Prototype of the IoT-Based Energy Monitoring and Control System, showcasing the integration of the microcontroller, current sensors, and relay card within the control unit, designed for real-time energy monitoring and remote control capabilities.

On the software side, the Arduino IDE was used to program the microcontroller with the necessary code to read the sensor data, perform calculations, and communicate with the Wi-Fi module. The ESP8266 was programmed using the AT command set to establish a connection to the Firebase database and transmit the processed data. A web-based user interface was developed using Google Charts, enabling users to visualize the energy consumption data in real-time and control the electrical systems remotely. The interface was designed to be user-friendly, with graphical representations of the data and easy-to-use controls for managing the energy systems.

C. Testing and Validation

Once the system was developed, extensive testing was conducted to ensure its accuracy, reliability, and functionality. The ACS712 current sensor was tested under various load conditions to verify its accuracy in measuring current. Different types of electrical devices, such as lights, fans, and air conditioners, were connected to the sensor to simulate real-world scenarios. The sensor readings were compared to those from a standard electrical meter to validate the accuracy of the measurements. The ESP8266 Wi-Fi module was tested for its ability to maintain a stable connection to the Firebase database and transmit data without significant delays or losses. The data transmission speed and reliability were critical factors in ensuring that the system could provide real-time monitoring capabilities.

The entire system was then tested in a controlled environment to identify and resolve any potential issues before deployment. The testing phase also included user testing of the web-based interface to ensure that it was intuitive and met the needs of the intended users. Feedback from users was used to refine the interface and improve the overall user experience.

D. Deployment

After successful testing, the system was deployed in selected rooms and buildings at Caraga State University. The deployment involved installing the ACS712 current sensors in the electrical circuits of these locations, configuring the Arduino Uno microcontrollers and ESP8266 Wi-Fi modules, and setting up the Firebase database for real-time data storage. The system was configured to monitor energy consumption continuously, with data being transmitted to the cloud database and made accessible through the web-based interface. Users were trained on how to use the interface to monitor energy usage and control the electrical systems remotely.

The deployment phase also included ongoing monitoring and maintenance of the system to ensure its continued reliability and effectiveness. Any issues that arose during the deployment were addressed promptly, and the system was adjusted as needed to optimize performance.

The methodology adopted in this study ensured that the IoT-based Energy Monitoring and Control System was developed, tested, and deployed successfully, providing a reliable and better solution for managing energy consumption at Caraga State University. Through careful design, rigorous testing, and systematic deployment, the system has demonstrated its potential to significantly reduce energy costs and improve energy efficiency in educational institutions..

III. RESULTS AND DISCUSSIONS

The implementation of the IoT-based Energy Monitoring and Control System at Caraga State University (CSU) provided valuable insights into the energy consumption patterns within the institution. The system's effectiveness was evaluated through extensive testing, real-time data analysis, and user feedback, leading to several key findings regarding the accuracy, reliability, and impact of the system on energy management. This section presents a detailed discussion of the results, supported by relevant figures and tables, to illustrate the system's performance and its implications for energy management at CSU.

1. System Performance and Accuracy

One of the primary objectives of this study was to assess the accuracy and reliability of the IoT-based system in monitoring energy consumption. The system's core component, the ACS712 current sensor, was rigorously tested under various load conditions to determine its accuracy in measuring current flow. Figure 3 below illustrates the calibration process of the ACS712 sensor, comparing the sensor readings with those obtained from a standard electrical meter.



Fig. 3: Calibration of ACS712 Current Sensor. No Load Current Reading of Digital Multimeter.

The results from the calibration tests showed that the ACS712 sensor provided readings with an error margin of $\pm 1.5\%$, which is within the expected range. This level of accuracy is sufficient for monitoring energy consumption in real-time and aligns with the requirements for effective energy management. The sensor's performance was consistent across different types of electrical loads, including lighting systems, air conditioners, and computing devices, as shown in Table 1.

TABLE II. ACCURACY OF ACS712 CURRENT SENSOR UNDER VARIOUS LOADS

Load Type	Standard Meter Reading (A)	ACS712 Sensor Reading (A)	Error (%)
Light Bulb (60W)	0.27	0.265	1.85
Air Conditioner	4.2	4.15	1.19
Laptop Charger	0.24	0.238	0.83
Electric Fan	0.78	0.774	0.77

The data in Table 2 confirms that the ACS712 sensor performs reliably across a range of load types, with minimal error. This consistency is critical for ensuring that the energy monitoring system provides accurate data, which is essential for making informed decisions about energy management.

2. Real-Time Monitoring and Data Collection

The IoT-based system's ability to collect and transmit real-time data on energy consumption was another critical aspect of the study. The ESP8266 Wi-Fi module played a central role in ensuring that data from the ACS712 sensor was transmitted to the Firebase database without significant delays or losses. Figure 2 presents a time series of data collected from one of the monitored rooms over a 24-hour period, highlighting the system's real-time monitoring capabilities.

To illustrate the system's effectiveness, Table 3 provides a summary of the total energy consumption for selected rooms before and after the implementation of the IoT-based system. The data shows a clear reduction in energy usage, demonstrating the system's potential for improving energy efficiency.

TABLE III. ENERGY CONSUMPTION BEFORE AND AFTER SYSTEM IMPLEMENTATION

Room	Energy Consumption Before (kWh)	Energy Consumption After (kWh)	Percentage Reduction (%)
Room A	85.4	72.3	15.3
Room B	92.1	77.6	15.8
Room C	78.5	67.2	14.3
Room D	105.3	89.1	15.4

The data in Table 2 shows a significant reduction in energy consumption across all monitored rooms, with percentage reductions ranging from 14.3% to 15.8%. These findings highlight the effectiveness of the IoT-based system in promoting energy efficiency through real-time monitoring and remote control capabilities.

3. Impact on Energy Costs and Savings

The implementation of the IoT-based Energy Monitoring and Control System at CSU had a measurable impact on energy costs. The reduction in energy consumption, as detailed in Table 2, translated directly into cost savings for the university. To quantify these savings, Table 3 provides an analysis of the electricity bills for the monitored rooms before and after the system was deployed.

TABLE IV. ELECTRICITY COST SAVINGS AFTER SYSTEM IMPLEMENTATION

Room	Cost Before Implementation (PHP)	Cost After Implementation (PHP)	Cost Reduction (PHP)
Room A	10,248	8,673	1,575
Room B	11,058	9,326	1,732
Room C	9,418	8,045	1,373
Room D	12,636	10,701	1,935

As shown in Table 3, the reduction in energy consumption led to significant cost savings, with reductions in electricity bills ranging from PHP 1,373 to PHP 1,935 per room. These savings demonstrate the financial benefits of implementing an IoT-based energy management system, particularly in environments with high energy consumption like universities.

4. Challenges and Limitations

While the IoT-based Energy Monitoring and Control System demonstrated significant benefits, the study also identified several challenges and limitations. One of the

primary challenges was the dependency on a stable internet connection. The ESP8266 Wi-Fi module relies on a continuous connection to transmit data to the Firebase database, and any disruption in connectivity could result in data loss or delays in implementing control commands.

Another limitation was the absence of voltage sensors in the system, which could have provided more accurate calculations of power consumption. The current system estimates power consumption based on current measurements alone, which, while effective, could be further improved with the inclusion of voltage data. Future iterations of the system should consider integrating voltage sensors to enhance the accuracy and reliability of the energy monitoring process.

Additionally, the system was deployed in a limited number of rooms, which, while sufficient for the purposes of this study, does not fully represent the potential impact of the system when scaled to cover the entire university campus. Future work should explore the scalability of the system and its potential to manage energy consumption across larger areas or multiple buildings.

7. Potential for Future Improvements

The findings from this study suggest several areas for future improvement and development of the IoT-based Energy Monitoring and Control System. Integrating voltage sensors into the system would allow for more precise calculations of power consumption, leading to even greater accuracy in monitoring and managing energy usage. Additionally, developing IoT-based magnetic contactors could improve the efficiency of the remote control feature, enabling faster and more reliable switching of electrical systems.

Expanding the system to include other forms of energy monitoring, such as water and gas usage, could provide a more comprehensive solution for resource management at CSU. This multi-faceted approach would allow the university to manage all of its utilities more effectively, leading to further cost savings and sustainability benefits.

Finally, exploring the use of machine learning algorithms to analyze the collected data could enhance the system's ability to predict energy usage patterns and optimize energy management strategies. By incorporating predictive analytics, the system could automatically adjust energy settings based on historical data and real-time conditions, further reducing energy consumption and costs.

IV. CONCLUSION

The IoT-based Energy Monitoring and Control System implemented at Caraga State University has proven to be an effective tool for managing energy consumption and reducing costs. The system's real-time monitoring and remote control capabilities offer significant advantages over traditional energy management methods, allowing for more proactive and efficient use of resources. The results of this study demonstrate the potential of IoT technology to revolutionize energy management in educational institutions, providing a model that can be scaled and adapted to meet the needs of other organizations.

The system's success at CSU underscores the importance of continuous innovation in energy management technologies. By addressing the challenges and limitations identified in this study and pursuing the

suggested improvements, future iterations of the system could achieve even greater efficiency and impact. The findings from this study contribute valuable insights to the growing field of IoT-based energy management and highlight the potential for these technologies to drive sustainability and cost savings in a variety of settings.

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